EECE 693 (Neural Networks / Deep Learning) – Project Proposal

Early Warning Prediction of Asthma Exacerbations Using Wearable and Digital-Health Data

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1 Problem Statement

Asthma exacerbations are clinically important events characterized by acute or subacute episodes of worsening respiratory symptoms and/or lung function deterioration, often requiring escalation of therapy to prevent serious outcomes [1, 2]. Because exacerbations can evolve over time rather than appearing instantly, early indicators may exist before a patient reaches severe deterioration.

In routine care, asthma status is frequently assessed using intermittent measures (e.g., symptom reporting, rescue inhaler use, and periodic clinic-based testing). This can delay recognition of worsening control in real-world settings, especially outside the clinic. Recent digital-health evidence suggests that exacerbation-related changes can be detected using "digital markers" derived from continuous or near-continuous monitoring, supporting the feasibility of earlier warning approaches [3].

Wearable devices provide an avenue for continuous, real-world collection of physiological and behavioral time series relevant to respiratory health, including respiration-related signals, heart rate, and activity. Prior work and reviews describe wearable vital-sign monitoring as a promising direction for asthma monitoring and management in everyday environments [4]. However, converting these multimodal streams into reliable early predictions remains challenging due to real-world measurement noise, missing data, and substantial inter-individual variability typical of wearable time-series in uncontrolled settings [5].

Therefore, a key gap remains: robust models that can learn from multimodal wearable sequences to forecast an upcoming asthma exacerbation with a clinically meaningful lead time. In this project, we formulate early warning as a supervised time-series prediction problem: using a fixed historical window of wearable signals (heart rate and activity, and respiration-related features when available), the model estimates the probability of an exacerbation occurring within a future prediction horizon. To ensure feasibility, we will begin with a baseline model using the most consistently available wearable streams and a primary horizon (e.g., 24 hours), then evaluate shorter horizons (e.g., 12 and 6 hours) and progressively incorporate additional sensor modalities. Model performance will be assessed using metrics appropriate for rare-event prediction (e.g., ROC-AUC/PR-AUC, sensitivity at a controlled false-alarm rate, and calibration), with the goal of providing reliable early warnings under real-world wearable data constraints.

2 Motivation

This project is motivated by the high clinical and societal burden of asthma and the potential impact of preventing severe exacerbations. The World Health Organization estimates that asthma affected 262 million people in 2019 and caused approximately 455,000 deaths worldwide [6]. Exacerbations are among the most consequential events in asthma care because they can lead to urgent visits, hospitalization, and life-threatening deterioration if not addressed early [2]. In addition to health impact, asthma imposes substantial economic costs; a U.S. CDC analysis estimated that asthma accounted for $81.9 billion in total annual costs (medical costs, mortality-related costs, and productivity losses) based on 2008–2013 data [7].

A second motivation is that outcomes can improve when worsening is recognized early and patients take timely, structured actions. Supported asthma self-management—particularly when it includes a written action plan and regular review—has been reported to reduce the risk of hospitalization in evidence-based summaries [8]. This indicates that earlier awareness of deteriorating control can translate into actionable steps that reduce escalation.

Finally, there is growing evidence that exacerbations may be preceded by measurable signals detectable via digital technologies. A 2024 systematic review identified multiple "digital markers" associated with asthma exacerbations and discussed their potential for enabling near real-time monitoring of asthma control [3]. Wearables are a practical platform for this, because they can continuously capture real-world signals relevant to respiratory status (including respiration-

related measures, heart rate, and activity) [4]. These points motivate developing machine learning models that can learn early-warning patterns from multimodal wearable sequences and forecast exacerbations with a clinically useful lead time.

3 Literature Review

Recent research on predicting asthma exacerbations/attacks using machine learning can be organized into three major directions: (i) ML risk modeling across diverse data sources (clinical/self-report, EHR, environment), (ii) short-term forecasting from home monitoring time series, and (iii) digital-marker and multimodal sensing approaches aiming for earlier and more personalized warnings.

3.1 Overall ML landscape and recurring limitations

A narrative review emphasizes that asthma exacerbations are difficult to predict due to complex and interacting drivers (clinical history, lung function, adherence, comorbidities, behaviors, and environmental exposure), and surveys common ML approaches used in the literature (e.g., logistic regression, tree-based methods, and deep learning) [9]. It also highlights practical issues including variability in outcome definitions and the need for stronger validation and generalizability.

3.2 Home-monitoring time series: baselines and false alarms under rare events

de Hond et al. compared classical and ML models for predicting severe exacerbations using time-series home monitoring signals and found that ML did not outperform logistic regression in their setting [10]. They also discuss the false-alarm issue that can arise in rare-event prediction.

3.3 Connected inhaler sensing and multimodal monitoring

Lugogo et al. reported predictive performance using connected inhaler signals and gradient boosting [11]. Tsang et al. introduced AAMOS-00, a multimodal dataset combining smartwatch sensing, smart inhaler data, smart peak-flow measurements, and questionnaires designed for asthma attack prediction [12]. A 2024 systematic review supports the feasibility of using digital markers for near real-time monitoring of asthma control [3], and the DIGIPREDICT protocol reflects current movement toward multimodal early warning systems [13].

3.4 Summary of gaps

Key gaps include robust modeling under rare-event conditions and real-world constraints (false alarms, calibration, generalization) and effective learning from noisy, incomplete, heterogeneous wearable time series [5, 10]. These motivate a baseline-first modeling plan.

4 Models to Be Used

We formulate the task as **supervised time-series risk prediction**: given a historical window of wearable signals (e.g., the past 6–24 hours), predict the probability of an asthma exacerbation occurring in a future horizon (e.g., the next 6–24 hours). We will start with the most consistently available wearable streams (heart rate + activity) and then add asthma-relevant modalities from AAMOS-00 (inhaler use, peak flow, questionnaires, and environment).

Because exacerbations are **rare events**, we will report **PR-AUC** in addition to ROC-AUC, evaluate **sensitivity at controlled false-alarm rates**, and assess **probability calibration**

(reliability curves / Brier score). We will also prefer **patient-wise splits** to avoid data leakage across the same individual.

4.1 Tier 1: Strong baselines (feature-based)

Regularized Logistic Regression (L1/L2) will be our primary interpretable baseline using window-level engineered features such as: mean/variance of HR, activity intensity distribution, step counts, simple trends (slope), and time-of-day/circadian indicators. Logistic regression is supported by evidence where it remained competitive in severe exacerbation prediction [10].

Gradient-Boosted Decision Trees (GBDT) (e.g., LightGBM/XGBoost) will serve as a high-performance tabular baseline on the same engineered features, and later on multimodal features once inhaler/PEF/questionnaire signals are added [15].

4.2 Tier 2: Deep learning on sequences (end-to-end)

We will evaluate sequence models that take the **minute-level (or resampled) time series** as input:

- **1D CNN / Temporal Convolutional Networks (TCN)** to learn local temporal patterns efficiently and handle long contexts with dilations [16].
- **RNNs (GRU/LSTM)** as standard physiological time-series baselines, tested with different window sizes (6h, 12h, 24h).

4.3 Tier 3: Transformers (advanced, if data supports it)

If the effective sample size supports stable training, we will evaluate Transformer-style models for longer temporal context and multi-horizon prediction (e.g., predicting risk for 6h/12h/24h horizons) [17, 18]. Transformers will be used only if they show real gains under patient-wise validation and rare-event metrics.

4.4 Instructor-required extensions (implemented concretely)

Regularization: dropout, weight decay, early stopping, and careful normalization for deep models (e.g., layer normalization for Transformers, batch normalization within CNN blocks when appropriate).
Ensembles: probability averaging across the best-performing models; optionally stacking (e.g., GBDT + deep model outputs) if it improves PR-AUC/calibration [19].
Transfer learning: optional transfer learning will be used only for an auxiliary model if we introduce an external dataset (e.g., a benchmark for respiration/HRV estimation). The resulting derived features would then be fused into the main predictor.
Architecture exploration: compare feature-based models vs end-to-end sequence models, and test alternative fusion strategies as modalities are added.
Multimodal learning: staged fusion (smartwatch-only → add inhaler/PEF → add questionnaires/environment). We will compare **early fusion** (concat features before the model) vs **late fusion** (separate subnetworks then combine probabilities/embeddings) and report ablations.

5 Dataset(s) Specifics

We will use a baseline-first dataset strategy: start from the most reliable continuous wearable streams for a feasible end-to-end pipeline, then progressively incorporate additional asthma-relevant modalities. The main dataset is AAMOS-00, which is explicitly designed for asthma attack/exacerbation prediction with connected devices [12].

5.1 Primary Dataset: AAMOS-00 (Asthma Monitoring Study)

AAMOS-00 combines smartwatch sensing with connected asthma monitoring signals, enabling realistic modeling under missingness and real-world adherence constraints [12].

5.1.1 Source and License

AAMOS-00 is publicly available via the University of Edinburgh DataShare repository under a Creative Commons Attribution (CC BY 4.0) license [14].

5.1.2 Study Size (as reported by the dataset paper)

Phase 2 includes 22 participants and provides 2,054 unique patient-days of data [12]. Data availability differs by modality (e.g., not all streams exist for all days), which we will explicitly quantify during exploratory analysis.

5.1.3 Outcome (Label) and leakage-safe setup

We will follow the outcome definition and labeling procedure described in the AAMOS-00 paper/documentation [12]. During EDA, we will reproduce the published labeling steps and clearly document the exact thresholds and combination rules used.

To avoid leakage, we will enforce **patient-wise data splits** (train/validation/test at the participant level). We will also verify that any feature engineering uses only information available **up to the prediction time**.

5.1.4 Wearable streams (dataset-faithful)

In the public AAMOS-00 release, the smartwatch stream provides minute-by-minute:

- heart rate,
- activity type,
- activity intensity,
- steps,

as described by Tsang et al. [12]. The public release does **not** include raw PPG waveforms; therefore, any respiration-from-PPG approach is considered only as an optional extension if a separate verified dataset is introduced.

5.1.5 Multimodal streams and staged integration plan

5.1.6 Windowing and sample construction (model inputs)

We will convert longitudinal data into supervised samples by choosing:

- a **history window** (e.g., 6h, 12h, 24h of wearable data),
- a **prediction horizon** (e.g., next 6h/12h/24h),
- and a **sampling stride** (e.g., produce a sample every 30–60 minutes).

Each sample will be labeled by whether an exacerbation occurs within the future horizon per the dataset definition [12].

Table 1: AAMOS-00 Streams and Planned Integration

Data Stream	Reported Coverage	What It Represents	Planned Phase
Smartwatch data	1,567 days	HR + activity type/intensity + steps (minute-level)	Baseline v1
Smart inhaler usage	694 days	Reliever use events (timestamps/counts)	Phase 2
Smart peak flow meter	1,099 days	PEF readings (morning/evening)	Phase 2
Daily questionnaires	1,583 days	Symptoms / control self-reports	Phase 3
Weekly questionnaires	324 weeks	Broader periodic assessment	Phase 3
Environmental context	Throughout	Weather/pollen/air quality via APIs	Phase 3

5.1.7 Missingness, imbalance, and evaluation plan

Real-world adherence produces missing segments and unequal modality coverage. We will:

- quantify missingness per stream and per participant,
- test simple imputation/forward-fill baselines for engineered features,
- and for deep models, compare masking strategies vs removal of heavily-missing windows.

Because exacerbations are rare, we will use **PR-AUC**, report **sensitivity at fixed false-alarm rates**, and include **calibration** checks. We will also include ablations comparing smartwatch-only vs added modalities to measure the incremental benefit of multimodal information.

5.2 Optional Supporting Dataset: Asthma Mobile Health Study (AMHS)

AMHS is a smartphone-based asthma study available via the Synapse platform [20]. If needed, it can serve as a backup source for symptom-related feature ideas and generalization checks, but AAMOS-00 remains our primary dataset due to its connected-device design [12].

6 Gantt Chart

This section outlines a tentative 12-week timeline and the expected contributions of the group members.

Expected contributions (equal split)

- **Bahaa Hamdan:** dataset setup and documentation, baseline feature engineering for smartwatch-only, and baseline evaluation (ROC-AUC/PR-AUC/calibration) with error analysis.
- **Tarek El Mourad:** preprocessing pipeline (windowing, resampling, missing-data handling), patient-wise split implementation, and Phase 2 integration (inhaler + peak flow) with ablation experiments.
- **Anthony Kanaan:** deep sequence models (1D CNN/TCN and GRU/LSTM; optional Transformer if feasible), hyperparameter tuning + regularization, and ensemble experiments; supports final report integration.

Table 2: Project Timeline (Weeks 1–12)

Task / Week	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Data acquisition & familiarization												
Preprocessing & dataset preparation												
Baseline model (smartwatch-only)												
Baseline evaluation & error analysis												
Phase 2 integration (inhaler + peak flow)												
Phase 2 training & evaluation												
Phase 3 integration (questionnaires + environment)												
Final model evaluation & ablation comparisons												
Report writing & final submission preparation												

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